

Lab 10 Study of IP Addressing and Subnetting.

Q:1. Find default subnet mask, network bits, hosts per subnet, no. of subnets, subnet number, 1st valid IP Address, last valid IP Address and broadcast address.

(i) 8.1.4.5/16

Class: A (1-126)

Default mask: 255.0.0.0

Given mask: 255.255.0.0

Network bits: 16, Host bits: 16

Host per subnet: $2^{16} - 2 = 65534$ (usable)

No. of subnets: borrowed bits = 8

No. of subnets = $2^8 = 256$ subnets

Subnet address = 8.1.0.0

1st valid host: 8.1.0.1

Last valid host: 8.1.255.254

Broadcast Address: 8.1.255.255

(ii) 130.4.102.1/24

Class: B

Default mask: 255.255.0.0

Given mask: 255.255.255.0

Network bits: 24, Host bits: 8

No. of bits borrowed: 8

No. of subnets: $2^8 = 256$

Network address: 130.4.102.0

1st valid host: 130.4.102.1

Last valid host: 130.4.102.254

Broadcast Address: 130.4.102.255

(iii) 130.4.102.1/22

class: B

default mask: 255.255.0.0

Given mask: 255.255.252.0

Network bits: 22, Host bits: 10

Hosts per subnet = $2^{10} - 2 = 1022$ (usable)No. of bits borrowed: $22 - 16 = 6$ No. of subnets = $2^6 = 64$ 1st valid address: 130.4.102.1

Last valid address: 130.4.102.254

Broadcast Address: 130.4.103.255

(iv) 199.1.1.100/27

Class: C

default mask: 255.255.255.0

Given mask: 255.255.255.224

Network bits: 27

Host bits: 5

Hosts per subnet = $2^5 - 2 = 30$ (usable)No. of subnets = $27 - 24 = 3 \Rightarrow 2^3 = 8$ subnets1st valid: 199.1.1.97

Last valid: 199.1.1.126

Broadcast: 199.1.1.127

Q:2 A host in a class C network has been assigned an IP address 192.168.17.9. Find the no. of addresses in the block, the first & the last address.

→ default mask : 255.255.255.0
 No. of address in block = $2^8 = 256$
 usable : 254

First Address : 192.168.17.0

Last Address : 192.168.17.254

Q:3 An address in block is given as 185.28.17.9. Find the no. of addresses in the block, the first & last address.

→ No. of addresses in block = $2^{16} = 65536$
 usable = 65534

First Address : 185.28.17.0

Last Address : 185.28.17.255

Q:4 A block of address is granted to small organization. we know that one of the address is 205.16.37.39/28. what is the first, last address, no. of addresses in a block?

First Address : 205.16.37.32

Last Address : 205.16.37.46

no. of addresses : $2^{(32-28)} = 2^4 = 16$

usable : 14

Q:5
= Subnet the IP address 216.21.5.0 into 30 hosts in each subnet. Find class, default mask.

→ class : C

default : 255.255.255.0

~~bit~~ $2^5 = 32$, so bit required = 5

new mask = $32 - 5 = 27$

bit borrowed = $8 - 5 = 3$

No. of subnets = $2^3 = 8$

8 subnets, block size = 32

216.21.5.0 - 216.21.5.31

216.21.5.32 - 216.21.5.63

216.21.5.64 - 216.21.5.95

216.21.5.96 - 216.21.5.127

216.21.5.128 - 216.21.5.159

216.21.5.160 - 216.21.5.191

216.21.5.192 - 216.21.5.223

216.21.5.224 - 216.21.5.255

Q:6
= Subnet the IP address 192.10.20.0 into 52 hosts in each subnet. Find class, default mask, bit borrowed, new subnet mask, no. of hosts & subnet, network ranges.

→ class : C

NEED hosts = 52, $2^6 = 64$

Host bits required = 6

New mask = $32 - 6 = 26$

Bits borrowed = $8 - 6 = 2$

No. of subnets = $2^2 = 4$

Host per subnet = $2^6 - 2 = 62$

192.10.20.0 - 192.10.20.63

192.10.20.64 - 192.10.20.127

192.10.20.128 - 192.10.20.191

192.10.20.192 - 192.10.20.255

Q:7
=

a) Device A: 172.16.17.30/20

Here we have to find the subnet mask for device A.

mask /20 $\Rightarrow$ 255.255.240.0

240 $\Rightarrow$ 4 bits third octet $\Rightarrow$ 11110000 = 240

Network address: 172.16.16.0/20

First valid: 172.16.16.1

Last valid: 172.16.31.254

Broadcast: 172.16.31.255

b) Device B: 172.16.28.15/20

mask /20 $\Rightarrow$ 255.255.240.0

third octet = 28

block size = $255 - 240 = 16$

Network address: 172.16.16.0/20

First valid: 172.16.16.1

Last valid: 172.16.31.254

Broadcast: 172.16.31.255